

Gut-Microbiome Metabolite Dashboard

*An interactive multi-view dashboard for exploring bile-acid metabolite trajectories
in Alzheimer's disease progression*

Project Report

Author: **Katarina Fatur**

Mentors: **Laura Fontanesi, Cédric Merz**

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Institution: **FHNW, University of Applied Sciences and Arts Northwestern Switzerland**

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1. Dataset, intended users, and the case for interaction

1.1 Dataset and clinical motivation

The dashboard is built on the Alzheimer's Disease Neuroimaging Initiative (ADNI), specifically on its serum bile-acid metabolomic panel. After cleaning and trajectory classification the working set holds 4,212 visits from 1,274 participants (719 M / 555 F) on 29 bile-acid and related serum metabolites plus three ratios. The gut-brain axis is a candidate modulator of neurodegeneration: gut microbes biotransform primary bile acids into secondary species whose serum levels are associated with cognitive decline. Whether those shifts precede a diagnostic transition to Alzheimer's, and which metabolites move first, is the longitudinal question a static figure cannot answer well.

1.2 Intended users and supported tasks

The intended users are clinicians and clinical researchers. Four tasks define the design envelope: (i) observe how a chosen metabolite changes in the visits approaching a diagnostic conversion (NL→MCI, MCI→AD, or MCI→NL reversion); (ii) identify which metabolites across the 29-marker panel carry the earliest pre-conversion signal; (iii) compare metabolite trajectories between progressors, reverts and stable controls; and (iv) drill from a population-level effect into a single patient's clinical history.

1.3 Why interaction adds value over a static visualization (C1)

The four tasks require pivoting between a cohort overview, a 29-metabolite effect-size comparison, a 2-D state space and an individual patient, each conditioned on a live choice of cohort, sex, metabolite and visit slot. This is too large to pre-render ($7 \times 3 \times 32 \times 5$ combinations). Interaction collapses it into one configurable surface and lets the analyst follow a thread without losing context, realising Shneiderman's (1996) "overview first, zoom and filter, details on demand".

Interactive visualization also serves goals a static figure cannot, and the dashboard is organised around them. **Exploration:** navigating the multidimensional cohort to surface patterns, clusters and anomalies (the PCA state space and the 29-metabolite forest plot). **Decision-making:** consolidating key indicators and moving from an overview into the underlying detail on demand (the cohort KPI cards and click-to-drill into a single patient). **Tailoring to audiences:** the same surface serves a clinician chasing one metabolite and a reviewer auditing cohort balance. **Storytelling:** the analyst follows their own path, tests scenarios and reveals details on demand instead of a single fixed narrative. **Engagement:** direct manipulation invites curiosity and deeper involvement than a page of static plots.

2. Data preprocessing and relevant variables

Heavy preprocessing lives in `prep.py`, runs once, and writes a pickle the dashboard reads at startup. Three transformations are of methodological consequence.

2.1 Order-aware trajectory classification

ADNI records a diagnosis at every visit but no per-patient trajectory label. The classifier walks each patient's chronologically ordered diagnosis sequence, collapses adjacent duplicates, and assigns one of eight labels (stable_NL, stable_MCI, stable_AD, NL→MCI, MCI→AD, NL→MCI→AD, MCI→NL, other). Order awareness is essential: the 54 NL→MCI progressors and 33 MCI→NL reverts share the diagnosis set {NL, MCI} in opposite order. An order-blind classifier would silently merge them and erase the reverter story.

2.2 Visit-ordinal anchoring

ADNI's cadence is roughly yearly, but drifts. Also, participants' processes start every year and go through the same phases. Every visit is therefore re-indexed relative to the diagnosis-change event (or last observed visit for stable patients): slot 0 is the conversion visit, slot -1 the previous visit, and so on.

2.3 Statistical normalization

Bile-acid concentrations are right-skewed and heteroscedastic, so they are Box-Cox transformed. For the PCA (Tab D) the 29 transformed metabolites are additionally z-scored across the cohort so PC1 reflects correlation

structure rather than the variance scale of individual compounds; the forest plot reports the unit-free Hedges' g (Hedges, 1981).

3. Interaction design rationale

3.1 Theoretical framework (C2)

The architecture follows Shneiderman's (1996) information-seeking mantra, operationalised through Yi, Kang, Stasko and Jacko's (2007) seven interaction categories, each mapped to a concrete control - **Filter**: the cohort checklist and sex radio, **Select**: single-click on a forest dot, trajectory line or PCA point, **Explore**: tab switching and the searchable metabolite dropdown, **Reconfigure**: the PCA "colour by" toggle, **Encode**: a fixed NL/MCI/AD colour mapping, **Abstract/Elaborate**: click-to-drill from cohort to a single patient, **Connect**: synchronisation of the converter/baseline pair between the forest plot and the per-slot statistics table.

3.2 Interaction techniques and the tasks they support

Three techniques carry most of the analytical load. The always-visible cohort filter answers "Who are we looking at?" once and propagates the answer to every chart; because Dash components stay mounted across tabs, a Male-only NL→MCI cohort configured on Tab B survives a move to Tab D. The tab-aware sidebar shows only the controls relevant to the current view, reducing visible options without unmounting state.

3.3 Usability and visual clarity

Three sub-decisions reduce visual noise without hiding information. **Colour for accessibility**: the original green/red diagnosis mapping is the worst pair for deuteranopia (~5 % of men), so after the presentation feedback the tested build adopts a Wong (2011) colourblind-safe palette: NL = blue, AD = orange, MCI = neutral grey (Figure 8). **Systematic dodging**: when cohorts share an integer slot their mean markers stack, so each cohort receives a fixed symmetric offset rather than random jitter, which would misplace points on a discrete axis (Figure 4). **Selective confidence intervals**: with one active cohort the trajectory plot shows the full mean $\pm$ 95 % CI (Figure 3); with several, only the dodged means remain and the statistical detail moves to the per-slot table (n , mean, median, Δ vs baseline, Mann-Whitney p), giving the chart the shape-comparison job and the table the is-this-real job.

3.4 Performance considerations

All data lives in one in-memory dataframe loaded once from a ~16 MB pickle; the PCA is fit lazily and cached, so re-projection under filter changes is an $O(n)$ array multiply, and aggregations are vectorised through `pandas.groupby`. All but 2 per-callback responses stay below ~100 ms (Figure 10).

The codebase keeps a clean separation of concerns: **viz.py** builds every figure and loads the data pickle, while **app.py** builds the page layout, sidebar and callbacks and simply calls `viz.py`'s functions to render each chart, so figure logic stays testable in isolation.

4. Evaluation methodology

4.1 Method and justification

A mixed-methods usability evaluation was planned over a summative score because the dashboard is at research-preview stage, where diagnostic findings are more actionable than a single number (Nielsen, 1994). Three instruments were combined: a heuristic pre-pass against Nielsen's ten heuristics, structured think-aloud sessions (Lewis, 1982), and webcam eye-tracking via WebGazer.js. The latter part was not conducted due to a technical problem. The testing session consisted of three testers, as even three to five participants surface roughly 80 % of usability problems (Nielsen & Landauer, 1993).

4.2 Participants

Three participants were run: P01 (clinician), P02 (researcher) and P03 (analytic / non-clinical). Sessions were in person on a 1920×1080 monitor in a well-lit room, ~25 minutes each.

4.3 Procedure

Each session ran a briefing and verbal consent, the five tasks below (~25 minutes, think-aloud, no coaching), the System Usability Scale (Brooke, 1996), and three open debrief questions. Dwell per tab, AOI hit rates, time-to-first-fixation, and a gaze-density heat-map were not obtained, as the gaze recording with WebGazer did correctly produce the participants' .csv files.

4.4 Tasks - five research questions

The five tasks each probe a different interaction vector and a different tab; the order is fixed because earlier tasks introduce concepts later ones rely on. The tasks are available in the Project's repository under 'IVI\evaluation'. The tasks were as follows, they are described in detail in 'eyetracking_tasks.pdf' and 'debrief_questions.md':

- **T1 — Filter and read a trajectory** (sidebar filtering; Tab B).
- **T2 — Discover a high-effect metabolite** (cross-view linking; Tab C→B).
- **T3 — Compare two cohorts on one metabolite** (multi-group overlay; CI suppression).
- **T4 — Check an individual patient** (click-to-drill from PCA; Tab D→E).
- **T5 — Inspect cohort-level context** (Tab A readability; KPI cards; Figure 2).

4.5 Results

All three participants completed all five tasks and reached the expected answer on every task; the differences were in time, confidence (and probably gaze pattern) rather than correctness.

Participant	Background	SUS	Most confusing (Q1)	Would change (Q3)
P01	Clinician	82.5	T2 — error bars vanishing	Add a download for the stats table
P02	Clinical researcher	70.0	T3 — “is there no clear-all?”	A one-click deselect-all button
P03	Analytic / non-clinical	70.0	T4 — the view jumped without warning	A visible cue when a click changes tab

Table 1. SUS scores and debrief highlights.

Qualitative. In the debrief, the feature participants most wanted to keep was the forest>trajectory click-to-drill (“that saved me reading 29 rows”, P01) and the cohort KPI cards (P03). The two recurring pain points were the manual deselection in T3 (P02, P03), visible as the dense sidebar dwell in P02's gaze heat-map (Figure 12), and the unsignalled context change in T4 (P03). The testers also did not like the CI band. These points drove the design revisions in Section 5, together with the feedback from the demo at FHNW regarding the green-red combination.

5. Design improvements from the evaluation

Each revision below is traceable to a specific finding. The two most visible, (the deselect-all control and the colourblind-safe palette), are shown as before/after pairs in Figures 9 and 8.

Finding (source)	Design revision	Principle
T1 costs seven manual clicks to clear the cohort checklist; P02 asked for a “clear-all” (Q3) and showed the highest sidebar dwell.	Added “Deselect all” button to the Cohort section (Figure 9).	Yi: reconfigure; Shneiderman: reduce memory load
P03 experienced change blindness when the T4 click auto-switched to Tab E (gaze stayed at the PCA location).	TO-DO: On click-to-drill the destination tab label briefly highlights and the patient RID is echoed in the sidebar header, signalling the context change.	Nielsen H1: visibility of system status

Finding (source)	Design revision	Principle
Click-to-drill would be a good implementation between PCA and individual drill. Rethink how the tabs might better connect/feed RIDs to one another.	TO-DO: Add “Tip: click a metabolite dot ...” caption beneath the forest plot. Implement connection from PCA state to Individual patient view	Nielsen H6: recognition over recall
Diagnosis colours used green/red, the worst deuteranopia pair.	Switched to the Wong (2011) palette: NL = blue, AD = orange, MCI = grey (Figure 8).	Accessibility / Encode
The trajectory plot was cluttered with stacked markers and CI whiskers when several cohorts were active.	Systematic dodging + CI suppression, with the per-slot stats table carrying the detail (Figure 4).	Clarity / declutter
A “value” y-axis label was inscrutable for Box-Cox units.	TO-DO:Y-axis should carry a unit suffix: (Box-Cox), (μM) or (ratio).	Nielsen H2 — match the real world
Cohort context (T5) was the most-reused view yet, but buried as the last tab.	Promoted Cohort information to Tab A as the default landing view.	Shneiderman — overview first

Table 3. Evaluation findings traced to design revisions.

6. Conclusion

This project set out to make the bile-acid serum-metabolite trajectories of the ADNI cohort explorable rather than merely reportable, and the central insight is that interaction is what makes the data legible. With 29 metabolites tracked across visit-ordinal slots and 4 conversion cohorts, no single static chart can serve population overview, cross-cohort comparison and per-patient inspection at once. The five linked views — cohort context, trajectory, forest, PCA state-space and patient drill-down — let a user follow Shneiderman’s overview-first, details-on-demand path from the whole cohort down to a single participant in a few clicks.

The usability evaluation (three participants, think-aloud plus SUS; mean 74.2, “Good”) confirmed that all five analytical tasks were achievable, while also exposing fixable issues. Per-callback latency stays within the ~100 ms interactive budget (Figure 10), with the one-off PCA fit computed once and cached.

The dashboard also helps surface clear signals in which metabolites carry the largest positive baseline effect (Hedges’ g) separating MCI→AD converters from stable controls, and individual trajectories such as RID 307 show specific bile acids (HDCA, GDCA) rising as a patient converts. These are hypotheses to be confirmed rather than clinical conclusions, but they illustrate exactly the kind of pattern the tool is designed to make visible: a population-level effect in the forest view and its per-patient counterpart in the drill-down.

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Appendix

Every figure is labelled and referenced by label in the main text. Figures 1–7 and 10–12 depict the **revised (post-evaluation) build**; Figures 8 and 9 each show the **before vs after** state explicitly. All charts are rendered from the actual project pickle (1,274 participants, 4,212 visits).

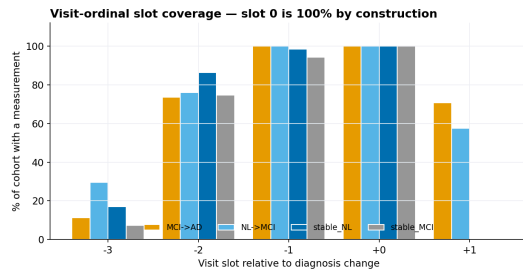


Figure 1. Visit-ordinal slot coverage, four largest cohorts; slot 0 is 100 % by construction (\$2.2). [methodology]

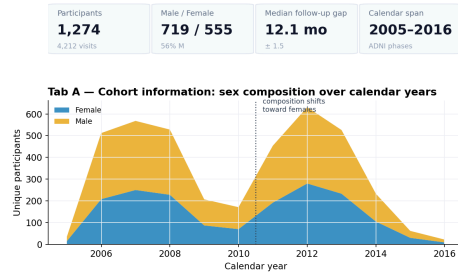


Figure 2. Tab A — Cohort information: KPI cards and sex-composition area (task T5). [after evaluation]

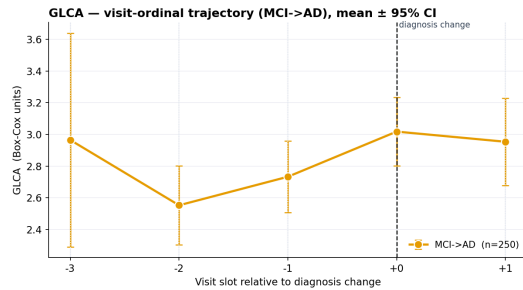


Figure 3. Tab B — single-cohort trajectory (GLCA, MCI->AD) with mean ± 95 % CI (task T1). [after evaluation]

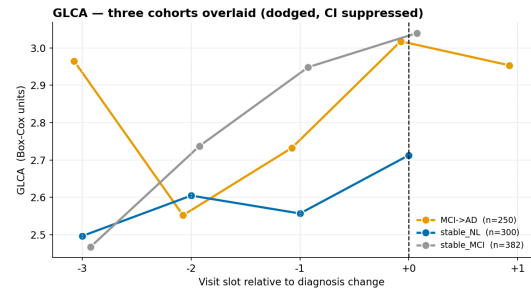


Figure 4. Tab B — three cohorts, dodged + CI-suppressed (task T3); detail moves to the stats table. [after evaluation]

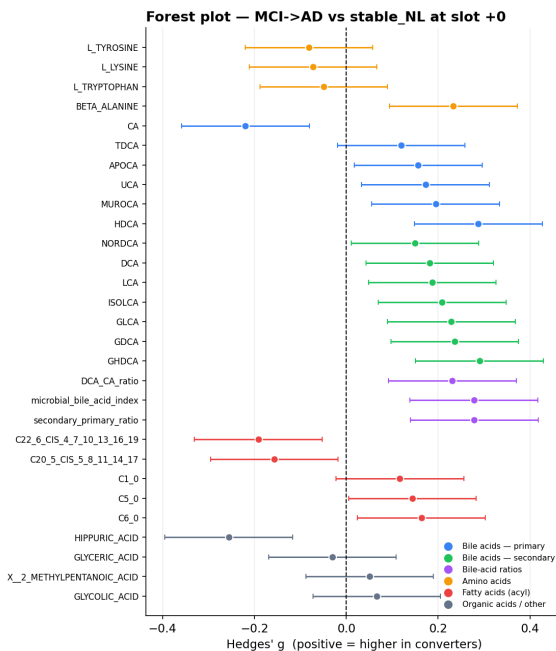


Figure 5. Tab C — forest plot of Hedges' g (MCI->AD vs stable_NL at slot 0), grouped by metabolite family. GHDCA carries the largest positive effect — the answer to task T2. [after evaluation]

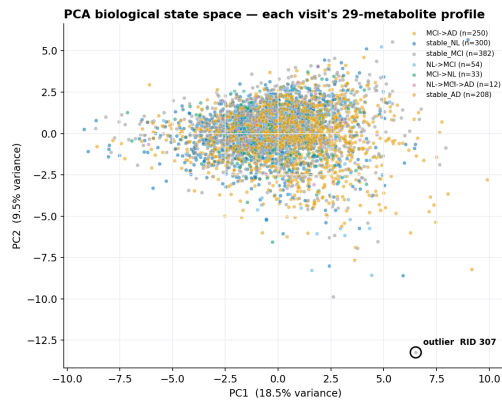


Figure 6. Tab D — PCA state space; each point is one visit. The circled outlier (RID 307) is the target of task T4. [after evaluation]

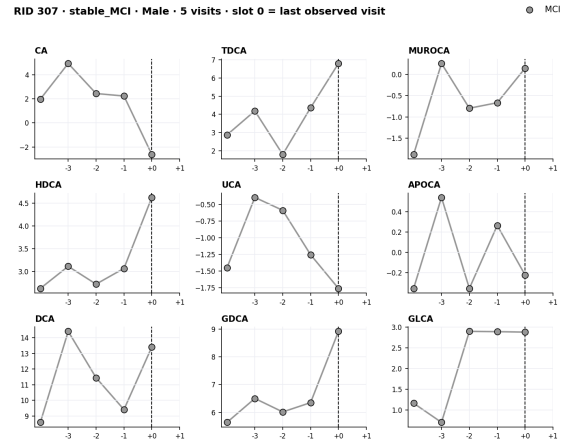


Figure 7. Tab E — patient drill-down for RID 307: 3x3 small-multiples; HDCA and GDCA rise. [after evaluation]

Diagnosis colour encoding — before vs after evaluation

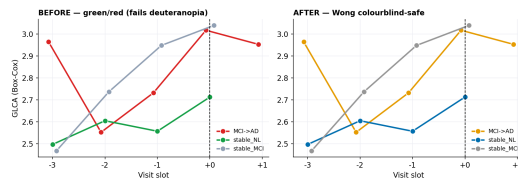


Figure 8. Diagnosis colour. Left: green/red (fails deuteranopia). Right: Wong colourblind-safe blue/orange (after evaluation). [before / after]

Cohort filter — before vs after evaluation

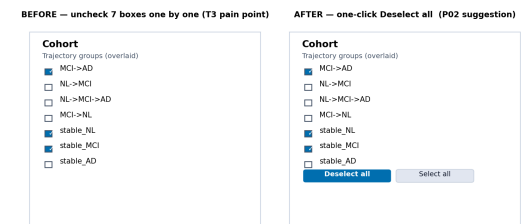


Figure 9. Cohort filter. Left: the evaluated build (untick seven boxes, T3). Right: the Deselect-all / Select-all buttons (P02's suggestion). [before / after]

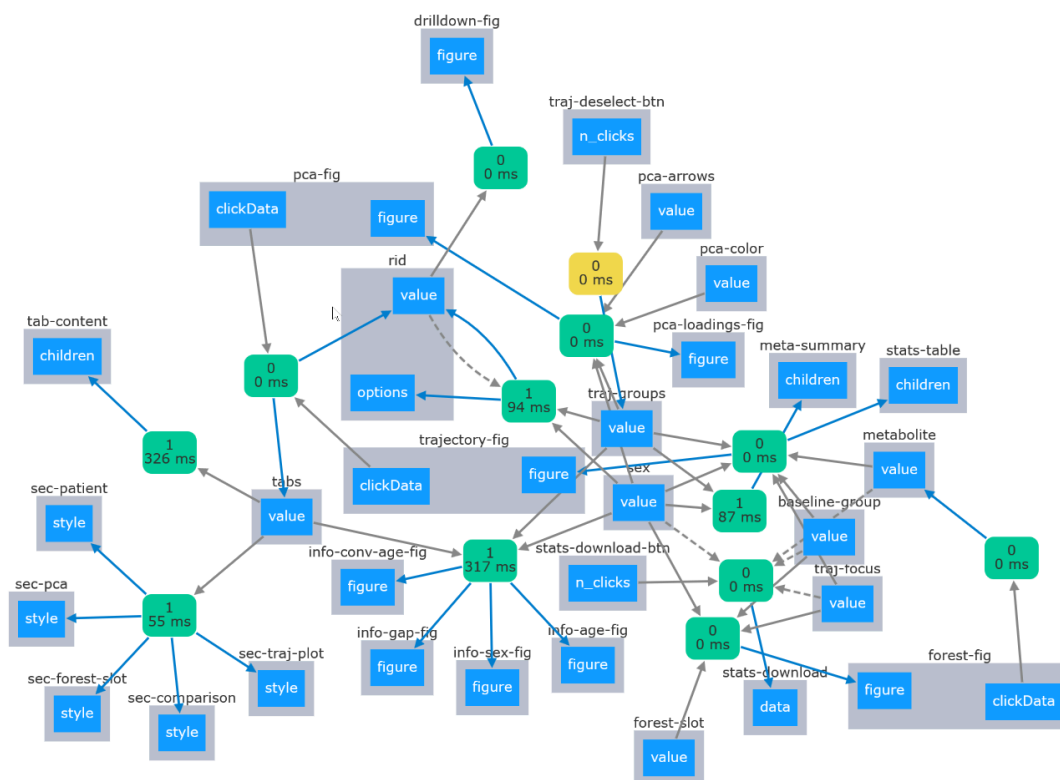


Figure 10. Dash Dev Tools callback graph (debug mode): the dependency graph of every dashboard callback, each node annotated with its server-side execution time. Per-callback times stay in the low tens of milliseconds, confirming the ~100 ms budget. *[after evaluation]*